

# Strategy comparison

Strategy comparison overview Scope - This document compares all strategy implementations currently in the repo. - It summarizes signal style, complexity, risk model, and notable implementation details. - Source details come from the per-strategy walkthrough files in docs/strategy\_walkthroughs.

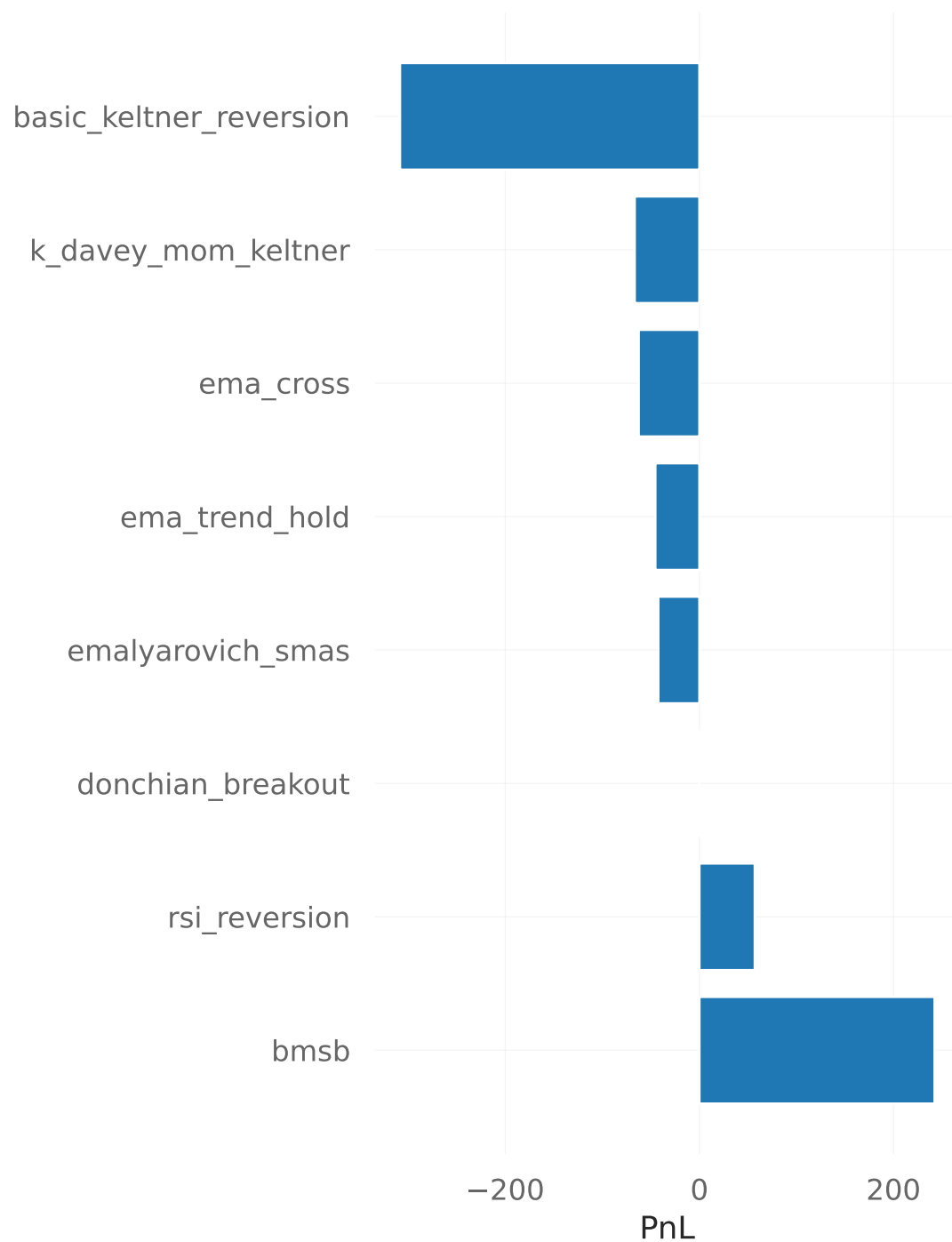
included 1) ema\_cross 2) rsi\_reversion 3) donchian\_breakout 4) ema\_trend\_hold 5) bmsb 6) emalyarovich\_smas 7) k\_davey\_mom\_keltner 8) basic\_keltner\_reversion Comparison matrix | Strategy | Signal style | Market bias | Shorts | Risk model | Position sizing | Complexity |

|                         |                                                           |         |    |                                                      |                                          |            |
|-------------------------|-----------------------------------------------------------|---------|----|------------------------------------------------------|------------------------------------------|------------|
| ema_cross               | Trend-following EMA crossover with trend filter           | Neutral | No | Percent SL/TP, optional ATR SL/TP, optional ADX gate | Backtester modes + optional position_pct | Medium     |
| rsi_reversion           | Mean reversion from RSI oversold to RSI recovery          | Neutral | No | Percent SL/TP only                                   | Backtester modes + optional position_pct | Low        |
| donchian_breakout       | Breakout above prior high, exit below prior low           | Bias    | No | Trend breakout                                       | No (default)                             | Low-Medium |
| ema_trend_hold          | above trend EMA, hold until close below trend EMA         | Bias    | No | Trend hold                                           | No (default)                             | Low-Medium |
| bmsb                    | Price over blended baseline plus rollover                 | Neutral | No | signal confirmation                                  | Trend with confirmation                  | Low        |
| emalyarovich_smas       | Pullback to fast SMA inside higher-timeframe trend        | Bias    | No | Backtester SL/TP inputs                              | Backtester modes + optional position_pct | Medium     |
| k_davey_mom_keltner     | Momentum + trend + Keltner thresholds, next-bar execution | Neutral | No | Multi-filter trend/momentum                          | Yes                                      | High       |
| basic_keltner_reversion | Revert from KC extremes back toward EMA centerline        | Neutral | No | Mean reversion                                       | Percent SL/TP only                       | Low-Medium |

Step-by-step method for reading any strategy in this repo 1) Data load and warmup - The runner pulls OHLCV with futures mode, no fixed percent SL/TP | Dynamic contracts or optional position\_pct | High 2) Signal generation - Strategy function returns a tuple: (signal, trigger). - Common signals are LONG, SHORT, EXIT, or None. 3) Signal execution - bt.on\_signal\_position transitions. - Pyramiding behavior and trigger fallback labels are handled in Backtester. Outputs - Every runner computes stats, equity curve, CSV, and optional QuantStats report. - Some have extra plot generation flags; behavior is not always identical across runners. Complexity ranking (highest to lowest) 1) k\_davey\_mom\_keltner 2) ema\_cross 3) bmsb 4) emalyarovich\_smas 5) donchian\_breakout 6) basic\_keltner\_reversion 7) rsi\_reversion 8) ema\_trend\_hold Why this ranking - k\_davey\_mom\_keltner combines multi-filter entry logic, next-bar pending actions, ATR stop updates, futures PnL, and contract sizing. - ema\_cross and bmsb add meaningful filter/trailing behavior beyond simple on-bar triggers. - rsi\_reversion and ema\_trend\_hold are straightforward single-indicator systems with entry/exit gates. Key cross-strategy caveats you should know 1) Timing semantics are not fully consistent. - Some runners compute signals on prior-bar slices and execute on current close; K. Davey queues for next-bar open. 2) CSV schema is reused across strategies. - Fields like ema\_fast/ema\_slow exist even for non-EMA-crossover systems (often set to None or repurposed). 3) Repeated entry signals can happen by design. - Several strategies emit LONG repeatedly while condition stays true; actual exits depend on BacktesterV2 position/pyramiding checks. 4) Exit generation differs by strategy type. - Some have explicit indicator exits. - Others rely heavily on BacktesterV2 stop logic once in position. Recommended usage by archetype - If you want the simplest baseline: ema\_trend\_hold or rsi\_reversion. - If you want breakout logic: donchian\_breakout. - If you want pullback trend entries: emalyarovich\_smas. - If you want advanced futures-style dynamic sizing: k\_davey\_mom\_keltner. - If you want mean-reversion logic: basic\_keltner\_reversion.

# Synthetic baseline comparison

Total net PnL



Win rate

